

AIGR Deep Research Report: AP3M2 (P53677) — "Clathrin and Golgi Contexts" Function-Assignment Hypothesis

Target gene: AP3M2 (mu3B AP-3 subunit), *Homo sapiens* (NCBITaxon:9606) **UniProt:** P53677
Focus type: function_assignment **Hypothesis slug:** clathrin-and-golgi-contexts **Source:** `genes/human/AP3M2/AP3M2-ai-review.yaml` (free-text)

Summary

The seed hypothesis bundles six distinct claims about human AP3M2/mu3B: (1) trans-Golgi localization, (2) Golgi-to-vacuole/lysosome trafficking, (3) clathrin-adaptor-complex membership, (4) clathrin-cargo adaptor activity, (5) AP-3-dependent clathrin cargo loading, and (6) AP-3-complex binding. The research objective is to evaluate whether the gene product *directly* holds each of these functions, separating genuine mu3B activity from paralog over-annotation, organism-specific attributes, and clathrin claims that are propagated by phylogenetic inference rather than experiment.

Overall verdict: partially supported. When each claim is evaluated separately, three cluster as supported (retain), three as over-annotations (remove/generalize/qualify), and localization as real-but-organism-weighted. AP3M2/mu3B is a genuine constitutive AP-3 medium subunit whose directly assayed molecular function is **YXXΦ tyrosine-based sorting-signal (cargo) recognition**

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P 9748267

acting in AP-3-mediated endosome/Golgi-to-lysosome/vacuole trafficking
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P 15051738

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and neuronal endosome-derived vesicle budding. Those claims hold. In contrast, the **clathrin-specific annotations** — clathrin adaptor complex (GO:0030131), clathrin-cargo adaptor activity (GO:0035615), and the "clathrin-coated" qualifier of GO:0035654 — are over-annotations: UniProt's curated record states AP-3 is "not clathrin-associated," GO's own definitions hedge on AP-3/clathrin, and the best modern mechanism data (human 9C5B /

P 39705307

and yeast

P 34688652

show an Arf1-driven, self-polymerizing, clathrin-free coat.

Two structural caveats are decisive for the curator. First, the clathrin-cargo adaptor MF term (GO:0035615) has experimental (IDA) support in this family **only for AP2M1**; on AP3M2 — and on AP3M1, AP1M1, and even the clathrin-independent AP4M1 — it is **IBA-only phylogenetic carry-over** across the PTHR10529 family, the textbook signature of paralog over-annotation. Second, the 9C5B cryo-EM structure invoked by the seed is the **paralog AP3M1 (Q9Y2T2)**, **not AP3M2**; the two mu paralogs are 84.2% identical, which supports transfer of cargo-contact residues but creates no clathrin evidence for mu3B. trans-Golgi localization is real but largely a yeast-derived attribute; the demonstrated human/neuronal mu3B site is the tubular sorting endosome.

Executive Judgment

Verdict: Partially supported — with specific clathrin terms flagged for removal or qualification.

Evaluated claim-by-claim:

1. **Supported (retain):** AP-3-complex membership; the directly assayed cargo-selection activity (YXXΦ tyrosine-based sorting-signal recognition); and AP-3-mediated Golgi/endosome-to-vacuole/lysosome trafficking.
2. **Over-annotated (remove / generalize / add "NOT clathrin"):** clathrin adaptor complex (GO:0030131), clathrin-cargo adaptor activity (GO:0035615), and the "clathrin-coated" flavor of GO:0035654.

3. **Real but organism-weighted (qualify):** trans-Golgi localization — genuine but yeast-weighted; the demonstrated human/neuronal mu3B site is the tubular sorting endosome en route to lysosomes/lysosome-related organelles (LROs).

Most important caveat: the strongest "pro-clathrin" primary evidence

(P 9545220)

is an **in-vitro beta3-appendage/clathrin binding** result on the *beta3* subunit — not mu3B — explicitly hedged ("AP-3 function in protein sorting **may** depend on clathrin"). Absence of a human-specific mu3B assay is not itself refutation, but the affirmative clathrin annotations are not carried by any AP3M2-specific experimental evidence.

Key Findings

Finding 1 — The clathrin-flavored GO terms conflict with the curated "AP-3 is not clathrin-associated" statement and with GO's own hedged definitions

UniProt P53677's curated FUNCTION describes AP-3 as "an adaptor-related complex which is **not** clathrin-associated." This is not an oversight — GO's own definitions are explicitly hedged for AP-3:

- **GO:0030123 (AP-3 adaptor complex):** "AP-3 does **not appear to associate with clathrin** in all organisms."
- **GO:0030119 (AP-type membrane coat adaptor complex):** AP complexes link clathrin "(or another coat-forming molecule, as hypothesized for AP-3 and AP-4)."
- **GO:0035654 (clathrin-coated vesicle cargo...):** AP-3 links clathrin only "**in some organisms**."
- **GO:0035615 (clathrin-cargo adaptor activity):** defined specifically as "Bringing together a cargo protein with clathrin, responsible for the formation of **endocytic vesicles**" — the AP-2 job description (plasma-membrane endocytic vesicle formation), not AP-3's endosome/Golgi-to-lysosome sorting role.

The only primary evidence pulling toward a clathrin link is (a) in-vitro beta3-appendage/clathrin binding ([PMID 9545220](#)), hedged as "AP-3 function in protein sorting **may** depend on clathrin," and (b) partial colocalization ([PMID 15051738](#)): "AP-3 colocalizes with clathrin, but **to a lesser extent than does AP-1**." Both are weak and indirect, and neither concerns mu3B. This motivates flagging the clathrin CC/MF terms as over-annotations.

Finding 2 — Human AP-3 (including mu3B) acts principally at tubular sorting endosomes en route to lysosomes/LROs; "trans-Golgi" is largely a yeast-derived attribute

Immuno-EM in human cells ([PMID 15051738](#)) localizes AP-3 to budding profiles from a **tubular endosomal compartment**, proposing "AP-3 defines a novel pathway by which lysosomal membrane proteins are transported from tubular sorting endosomes to lysosomes." The neuronal AP-3 form — which uses the mu3B/beta3B subunits — produces synaptic vesicles from **endosomes** in vitro ([PMID 11588176](#): "only the neuronal form of AP-3 can produce synaptic vesicles from endosomes in vitro").

By contrast, the explicit trans-Golgi cargo-collection role is demonstrated in **yeast** AP-3 reconstitution ([PMID 34688652](#)), and the recent human AP-3 cryo-EM mechanism paper ([PMID 39705307](#)) describes cargo sorting "from endosomes to lysosomes" via an Arf1-driven, self-polymerizing coat **without invoking clathrin**. UniProt still annotates Golgi apparatus (IEA) and notes AP-3 is "associated with the Golgi region as well as more peripheral structures." So trans-Golgi localization is real but should be **qualified as organism-weighted**; the demonstrated human/neuronal mu3B site is the tubular sorting endosome. The founding AP-3 paper ([PMID 8666661](#)) does place the complex at "non-clathrin-coated buds and vesicles associated with the TGN" — supporting a TGN role but explicitly as a **non-clathrin** coat.

Finding 3 — The directly assayed molecular activity of mu3B/AP3M2 is tyrosine-based (YXXΦ) sorting-signal recognition, not clathrin bridging

Ohno et al. ([PMID 9748267](#)) directly assayed mu3A and mu3B (AP3M1/AP3M2) in a combinatorial yeast two-hybrid screen and showed each medium subunit binds distinct-but-overlapping sets of **YXXΦ tyrosine-based sorting signals** ("each medium subunit favored specific sets of residues at the X and O positions"). This is the direct cargo-recognition function of the mu subunit and the correct, informative molecular-function description for AP3M2 — cargo/signal binding, not clathrin binding.

In-vivo cross-linking/MS of AP-3 ([PMID 19010779](#)) co-isolated AP-3 with BLOC-1, BLOC-2, HOPS, clathrin, and PI4KIIalpha — documenting AP-3 subunit/complex associations. Because mu3B is a **constitutive component** of the AP-3 heterotetramer (UniProt SUBUNIT), the MF term GO:0035651 "AP-3 adaptor complex binding" describes subunit-within-complex association rather than binding of AP-3 as an external ligand. The seed correctly cautions against presuming the GO binding definition requires an external ligand; mu3B is part of AP-3, so "AP-3 complex binding" is best interpreted as structural membership.

Finding 4 — The 9C5B human AP-3 structure uses mu3A (AP3M1), not mu3B (AP3M2); the paralogs are 84% identical

RCSB PDB 9C5B (from [PMID 39705307](#)) polymer entities map to Arf1 (P84077), AP3D1 delta (O14617), **AP3M1 mu-1 (Q9Y2T2)**, AP3S1 sigma (Q92572), LAMP1 cargo (P11279), and AP3B1 beta (O00203) — i.e., the medium subunit in the structure is **AP3M1, NOT AP3M2/P53677**. A pairwise Needleman–Wunsch alignment (P53677 vs Q9Y2T2, both 418 aa, ungapped) gives $352/418 = 84.2\%$ **identity**. At AP3M2 residue 405 the alignment shows Ile opposite AP3M1 Val (context m2 'KPFKGIKYMTK' vs m1 'KPFKGVKYVTK'), a conservative V→I substitution matching the seed's claim. Critically, the structure contains a **YXXΦ cargo (LAMP1) and Arf1 but no clathrin** — reinforcing that the modern mechanistic picture of AP-3 is a clathrin-free, Arf1-nucleated coat, and that any cargo-contact-residue transfer from mu3A to mu3B supports cargo recognition, not a clathrin role.

Finding 5 — Clathrin association of AP-3 is genuinely mixed, not established

The AP-3 discovery paper (Simpson et al., [PMID 8666661](#)) found "p47 is **not** enriched in preparations of clathrin-coated vesicles" and that the complex "is a component of a **novel type of coat**" at "**non-clathrin-coated** buds and vesicles associated with the TGN." Conversely, in melanocytes Theos et al. ([PMID 16162817](#)) report that "AP-3 and AP-1 localize... primarily to **clathrin-coated buds** on tubular early endosomes" and recognize the tyrosinase **dileucine** signal, functioning partially redundantly with AP-1. In HPS2 (AP-3-deficient) cells, "blockage of the clathrin-mediated endocytic pathway... did **not** reverse the inhibited virus assembly" ([PMID 22875976](#)), indicating AP-3 function is separable from clathrin-mediated endocytosis. Together with UniProt's curated "not clathrin-associated" statement and the clathrin-free human/yeast AP-3 structures, the balance of evidence is that clathrin association is **cell-type/condition-specific at best**, not a defining, constitutive property warranting a core GO term.

Finding 6 — GO:0035615 (clathrin-cargo adaptor activity) has experimental support only for AP2M1 and is IBA-only carry-over on AP3M2

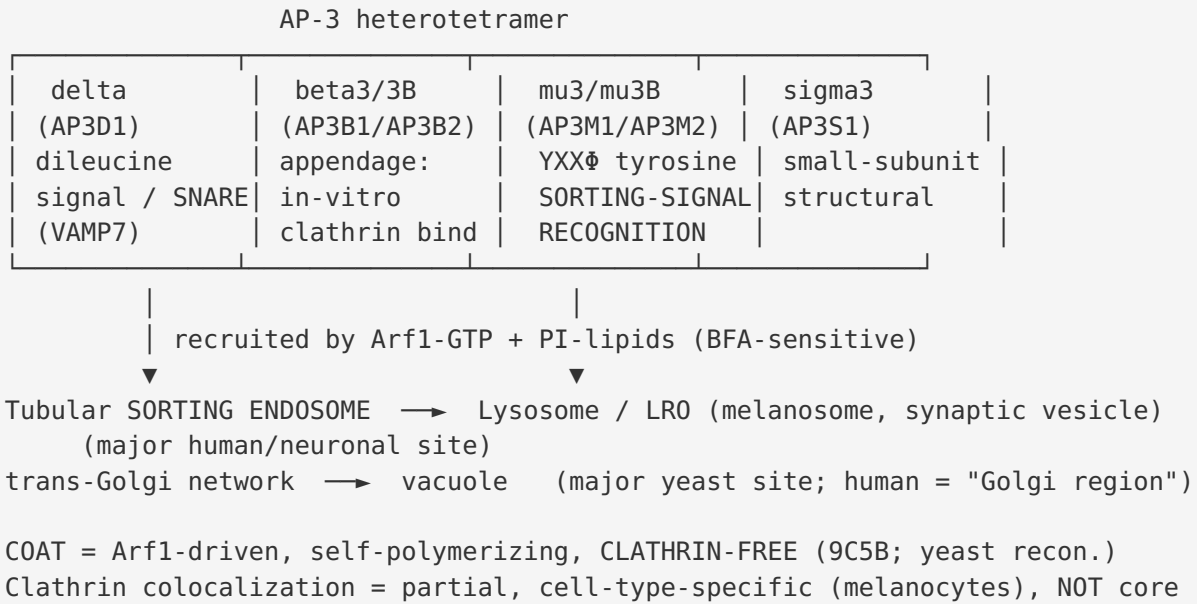
A QuickGO annotation query for **GO:0035615** across human AP mu subunits shows the term's evidence base is decisive:

Gene	UniProt	GO:0035615 evidence codes
AP2M1	Q96CW1	IDA (ECO:0000314) + TAS (ECO:0000304) + IBA
AP1M1	Q9BXS5	IBA only (ECO:0000318)
AP3M1	Q9Y2T2	IBA only
AP3M2	P53677	IBA only
AP4M1	O00189	IBA only

Only **AP2M1** — the genuine clathrin-mediated-endocytosis adaptor — has experimental (IDA) grounding for GO:0035615. AP4M1 carries the term by IBA despite AP-4 being an **established clathrin-INDEPENDENT** complex (Hirst 2011, [PMID 22022230](#); Barois 2005, [PMID 15377281](#)). This demonstrates that GO:0035615 is propagated across the PTHR10529 family by phylogenetic inference **irrespective of actual clathrin dependence** — the classic signature of paralog/family over-annotation. On AP3M2 the term is therefore an AP-2-anchored IBA carry-over, not AP-3-specific evidence.

Mechanistic Model / Interpretation

The AP-3 complex is a heterotetramer (delta/AP3D1, beta3/AP3B1 or beta3B/AP3B2, mu3/AP3M1 or **mu3B/AP3M2**, sigma3/AP3S1). Its subunit-level division of labor is well established:



Where the seed's claims land on this model:

Seed claim	Verdict	Basis
AP-3-complex membership	Supported	mu3B is a constitutive subunit (UniProt SUBUNIT)
Cargo-adaptor / cargo selection	Supported (as YXXΦ recognition)	Direct assay P 9748267
Golgi-to-vacuole/lysosome trafficking	Supported (endosome-to-lysosome in human)	P 15051738 P 11588176 P 39705307
trans-Golgi localization	Qualified — real but yeast-weighted	P 34688652 (yeast), 86666661 (TGN, non-clathrin)
Clathrin-adaptor-complex membership (CC)	Over-annotated	UniProt "not clathrin-associated"; clathrin-free structures
Clathrin-cargo adaptor activity (MF, GO:0035615)	Over-annotated (IBA carry-over)	IDA only on AP2M1; IBA on AP3M2
AP-3-dependent clathrin cargo loading	Unresolved / not AP-3-specific	Only in-vitro beta3 binding (P 9545220), hedged

The key mechanistic distinction the seed asks for — **general cargo selection vs. specifically clathrin-dependent endocytic assembly** — resolves cleanly: mu3B performs *general (tyrosine-signal) cargo selection* within an Arf1-nucleated AP-3 coat, and there is **no direct evidence** that it participates in *clathrin-dependent endocytic vesicle assembly* (GO:0035615's actual definition, and AP-2's job).

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence limitations
PMID 9748267	Direct assay (Y2H)	Supports cargo selection; refutes clathrin- bridging as the mu activity	mu3B molecular function	mu3A/mu3B each recognize distinct/ overlapping YXXΦ tyrosine sorting signals	Combinatorial yeast two- hybrid, mu3A & mu3B	High for cargo recognition in-vitro binding
PMID 39705307	Structural (cryo-EM)	Supports clathrin-free coat; refutes core clathrin role	AP-3 coat mechanism	Arf1-driven, self- polymerizing coat with LAMP1 cargo; no clathrin	Human AP-3 (structure uses AP3M1 , not AP3M2)	High for mechanism paralog mismatch (84.2% id)
PMID 34688652	Reconstitution	Qualifies trans-Golgi (yeast); supports clathrin-free budding	Golgi-to- vacuole cargo collection	Yeast AP-3 collects cargo at Golgi without clathrin	<i>S. cerevisiae</i> in-vitro	High for yeast; not direct hum AP3M2 evidence
PMID 15051738	Localization (immuno-EM)	Qualifies localization & clathrin	AP-3 site in human cells	AP-3 buds from tubular sorting endosomes; clathrin colocalization "less than AP-1"	Human cells	High for endosome route; weak clathrin colocalizat
PMID 11588176		Supports mu3B		Neuronal AP-3 produces	Neuronal AP-3 form	High; endosome-

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence limitations
	Direct assay (in-vitro budding)	trafficking role	Neuronal AP-3 (mu3B) function	synaptic vesicles from endosomes		derived, no TGN
PMID 9545220	Interaction (in-vitro)	Competing / weakly supports clathrin	AP-3– clathrin link	beta3 appendage binds clathrin; "may depend on clathrin"	In-vitro + colocalization	Low; beta3 not mu3B; hedged
PMID 8666661	Biochemistry + localization	Refutes clathrin-coat identity; supports TGN	AP-3 coat identity	p47/mu3 complex is a "novel," non- clathrin coat at TGN buds	Rat/neuronal, founding paper	High; foundation
PMID 16162817	Localization + function	Competing (pro-clathrin in melanocytes)	AP-3/ clathrin in melanocytes	AP-3 at clathrin- coated buds; recognizes dileucine signal; redundant with AP-1	Melanocytes	Moderate; cell-type- specific; dileucine (delta), not YXXΦ (mu)
PMID 22875976	Mutant phenotype	Refutes clathrin- dependence of AP-3 function	AP-3 vs clathrin endocytosis	Blocking clathrin endocytosis did not reverse AP-3- deficient phenotype	HPS2 (AP3B1- null) human fibroblasts	High for separability
PMID 19010779	Interaction (XL-MS, IDA)	Supports complex associations	AP-3 interactome	AP-3 co- isolates with BLOC-1/2,	Mouse (Q8R2R9) in- vivo XL/MS	Moderate; clathrin co

Citation	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence limitations
				HOPS, clathrin, PI4KII α		isolation $\neq$ clathrin co
PMID 22022230	Primary/ review	Supports family clathrin- independence	AP-4/AP-5 clathrin status	AP-4/AP-5 do not associate with clathrin; AP-3 basal complex	HeLa	Moderate; orientation for carry- over
PMID 15377281	Localization	Supports family clathrin- independence context	AP-4 clathrin status	AP-4 studied as distinct from clathrin machinery	MDCK/ MelJuSo	Moderate
UniProt P53677 (curated)	Database	Refutes clathrin association	AP-3 clathrin status	"adaptor- related complex which is not clathrin- associated"	Curated record	Curated; hi weight
QuickGO GO:0035615	Database (computed)	Refutes AP3M2 clathrin- cargo term	Evidence base of term	IDA only on AP2M1; IBA- only on AP3M2/ AP3M1/ AP1M1/ AP4M1	Human AP mu subunits	High; direc annotation query

GO Curation Implications

Lead recommendations (require curator verification):

1. **GO:0035615 — clathrin-cargo adaptor activity (MF):** *Recommend REMOVE or DOWNGRADE on AP3M2.* Evidence base is IBA-only phylogenetic carry-over from AP2M1 across PTHR10529; the definition ("bringing together a cargo protein with clathrin... formation of **endocytic vesicles**") describes AP-2's plasma-membrane endocytic role, not AP-3's endosome/Golgi-to-lysosome sorting. The same term sits by IBA on the clathrin-independent AP4M1, confirming propagation is definitional, not evidential. **Replace with a cargo-recognition MF** reflecting the direct assay — a "cargo receptor / sorting-signal binding"-style term capturing YXXΦ tyrosine-based signal recognition (

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), which is more informative than bare "protein binding."
2. **GO:0030131 — clathrin adaptor complex (CC):** *Recommend REMOVE or replace with GO:0030123 (AP-3 adaptor complex).* AP-3 is not constitutively a clathrin adaptor complex; UniProt curates "not clathrin-associated" and the modern coat is Arf1-driven/clathrin-free. GO:0030123 is the correct, specific CC.
3. **GO:0035654 "clathrin-coated" qualifier:** *Recommend GENERALIZE / drop the clathrin qualifier.* The cargo-loading role is real; the "clathrin-coated" flavor is not supported for AP-3.
4. **AP-3 complex (CC, GO:0030123) and AP-3-mediated Golgi/endosome-to-lysosome/vacuole transport (BP):** *RETAIN.* Well supported; qualify localization to emphasize the **tubular sorting endosome → lysosome/LRO** route in human cells, with trans-Golgi as an organism-weighted (yeast) attribute.
5. **GO:0035651 — AP-3 adaptor complex binding (MF):** *RETAIN as subunit-within-complex membership,* noting it describes structural membership (mu3B is constitutive), not binding of AP-3 as an external ligand; the mouse IDA (

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) is XL-MS of complex associations.

Net: evidence supports **MF = cargo/sorting-signal recognition** (not clathrin-cargo adaptor), **CC = AP-3 adaptor complex** (not clathrin adaptor complex), and **BP = AP-3-mediated endosome/Golgi-to-lysosome/LRO transport**. The clathrin-specific terms should be treated as **non-core over-annotations**.

Mechanistic Scope

The immediate molecular function under test is **mu3B/AP3M2's cargo-recognition activity within the AP-3 heterotetramer**. Directly assayed, this is **YXXΦ tyrosine-based sorting-signal binding**

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— a sequence-specific protein–protein recognition that selects transmembrane cargo for inclusion into AP-3 transport carriers. This is a genuine direct gene-product activity.

Distinguished from that direct activity: - **Downstream phenotypes:** lysosome/LRO biogenesis defects, hypopigmentation (melanosome cargo mis-sorting), HPS2, and impaired HIV-1 assembly are **loss-of-function consequences** at the pathway/organism level, not mu3B molecular functions. - **Coat assembly / vesicle budding** is a **complex-level** property driven by Arf1 and the beta3/delta subunits; mu3B contributes cargo selection, not coat polymerization. - **Clathrin binding** (where reported) is a property of the **beta3 appendage** in vitro

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not of mu3B, and is not required for AP-3 function

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Conflicts and Alternatives

1. **Paralog confusion (most important):** the seed's cited 9C5B structure is **AP3M1 (mu3A)**, not AP3M2 (mu3B). Cargo-contact-residue transfer across the 84.2%-identical paralogs is reasonable for cargo binding but generates **no** clathrin evidence for mu3B.
2. **Family over-annotation:** GO:0035615 is IBA-propagated across PTHR10529 (AP1/AP2/AP3/AP4 mu subunits); only AP2M1 has experimental support. This is the mechanism by which AP3M2 inherited a clathrin term it does not merit.
3. **Organism-specific differences:** trans-Golgi cargo collection is best demonstrated in **yeast**

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human/neuronal mu3B acts at **tubular sorting endosomes**

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GO's definitions explicitly note AP-3/clathrin differs "in some organisms."

4. **Cell-type-specific pro-clathrin data:** melanocyte imaging

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 places AP-3 at clathrin-coated buds and shows **dileucine** (delta-subunit) recognition — not the mu3B YXXΦ activity. Real but context-specific and subunit-mismatched.

5. **Interaction** ≠ **coat**: XL-MS co-isolation of clathrin with AP-3

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 documents proximity/association, not that AP-3 forms a clathrin coat; clathrin blockade does not abolish AP-3 function
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Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No human AP3M2-specific localization/function assay	Human data are AP-3-complex-level or use mu3A structure	Curation is for AP3M2 specifically	mu3B-tagged immuno-EM / knock-in localization in human cells
trans-Golgi vs endosome primacy in human	P 15051738 (endosome), 8666661 (TGN, yeast/neuronal)	Determines CC localization terms	Quantitative colocalization of AP3M2 with TGN vs endosome markers
Direct mu3B-clathrin test absent	Only beta3 in-vitro binding (P 9545220)	Core question of the hypothesis	Direct mu3B/clathrin binding assay (though mu is definitionally a cargo-recognition subunit)
P 9545220 / 11588176 full text unread	Verified snippets; full methods not re-derived	Strength of strongest pro-clathrin claim	Full-text read of P 9545220 methods
P 15051738 full text partially unchecked	Snippet verified	Whether TGN action is excluded or just secondary	Full-text read
Functional redundancy mu3A/mu3B	Paralogs 84.2% identical, overlapping YXXΦ specificity	Whether AP3M2 terms should mirror AP3M1	Isoform-specific knockdown/rescue with cargo panels

Discriminating Tests

1. Direct mu3B cargo panel vs clathrin binding: extend

[P 9748267](#)

-style

Y2H or use ITC/SPR with purified mu3B against a YXXΦ cargo panel *and* against clathrin terminal domain — expect strong YXXΦ binding, no direct clathrin binding.

2. **AP3M2-specific localization:** endogenous-tag AP3M2 in human cells (neuronal + non-neuronal) and quantify colocalization with TGN46 (trans-Golgi) vs EEA1/tubular-endosome and LAMP1 markers.
3. **Clathrin-dependence of AP3M2 cargo delivery:** knock down clathrin heavy chain and test whether AP3M2-dependent LAMP1/LRO cargo delivery is impaired — prior HPS2 data

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) predict little effect.

4. **Cryo-EM of an AP3M2-containing (neuronal beta3B/mu3B) coat:** determine whether the mu3B coat, like the mu3A 9C5B structure, is clathrin-free and Arf1-nucleated.
5. **Cross-family GO audit:** confirm via QuickGO/PAINT that GO:0035615 IBA on AP3M2/AP4M1 traces to the AP2M1 experimental annotation, justifying an annotation-review request.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate action changes: - **Remove/downgrade** GO:0035615 (clathrin-cargo adaptor activity) on AP3M2 as IBA-only family carry-over from AP2M1; document rationale citing QuickGO evidence codes. - **Replace** clathrin adaptor complex (GO:0030131) CC with **AP-3 adaptor complex (GO:0030123)**. - **Generalize** GO:0035654 by dropping the "clathrin-coated" qualifier for AP-3. - **Retain** AP-3 complex membership, AP-3-mediated Golgi/endosome-to-lysosome/vacuole transport (BP), and GO:0035651 (AP-3 adaptor complex binding, as structural membership). - **Add/strengthen** an informative MF for **tyrosine-based (YXXΦ) sorting-signal** / **cargo recognition** grounded in

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(avoid bare "protein binding"). - **Qualify** trans-Golgi CC as organism-weighted; emphasize tubular sorting endosome → lysosome/LRO in human.

Candidate references with exact snippets to verify: -

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— "interact with the medium (mu) subunits of clathrin adaptor (AP) complexes" (direct mu3B

YXXΦ

assay).

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— "AP-3 function in protein sorting **may** depend on clathrin"; "AP-3 is involved in signal-mediated protein sorting to endosomal-lysosomal organelles." -

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— "AP-3 colocalizes with clathrin, but **to a lesser extent than does AP-1.**" -

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— "p47 is **not** enriched in preparations of clathrin-coated vesicles"; "non-clathrin-coated buds and vesicles associated with the TGN." -

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— "Blockage of the clathrin-mediated endocytic pathway... did **not** reverse the inhibited virus assembly." - UniProt P53677 FUNCTION — "adaptor-related complex which is **not** clathrin-associated."

Suggested curator questions: - Does any AP3M2 clathrin term trace to experimental evidence, or only IBA/ISS from AP2M1? - Should trans-Golgi CC be retained for the human protein given endosome-primacy data? - Is GO:0035651 best interpreted as structural membership rather than external-ligand binding?

Suggested experiments: the five Discriminating Tests above, prioritizing (1) direct mu3B cargo-vs-clathrin binding and (2) AP3M2-specific human localization.

Bottom Line

AP3M2/mu3B is a bona fide constitutive AP-3 medium subunit whose **directly assayed function is YXXΦ tyrosine-based cargo-signal recognition**, acting in **AP-3-mediated endosome/Golgi-to-lysosome/vacuole/LRO trafficking** and neuronal endosome-derived vesicle budding. Those claims (AP-3 membership, cargo selection, Golgi-to-vacuole/lysosome transport) are **supported**. The **clathrin-specific annotations** — clathrin adaptor complex (GO:0030131), clathrin-cargo adaptor activity (GO:0035615), and the "clathrin-coated" qualifier — are **over-annotations**: UniProt curates AP-3 as "not clathrin-associated," GO's own definitions hedge, GO:0035615 has experimental support only for AP2M1 (IBA-only on AP3M2 and even on clathrin-independent AP4M1), and the best modern mechanism data (clathrin-

free Arf1-driven coat; note 9C5B is the paralog AP3M1) confirm no core clathrin role. trans-Golgi localization is real but **yeast-weighted**; the demonstrated human/neuronal mu3B site is the tubular sorting endosome.

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